Unit 6, Lesson 4: Connecting One-Variable Inequalities to Two-Variable Inequalities

Benchmarks

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.

MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

Learning Target

- [ ] I can connect the solutions to one-variable inequalities to the solutions of two-variable inequalities using graphs.

6.4.1 Warm-Up: History of Mathematics

René Descartes (1596-1650) was a French philosopher and mathematician. He is credited with the invention of the modern coordinate system, which is also called the Cartesian coordinate system — named after him. As the story goes, Descartes was sick in bed one day and saw a fly crawling on the ceiling. He determined that he could use just two numbers, one for the fly's horizontal position and one for its vertical position, to describe the fly's location at any point.

1. As the story goes, how did the fly help Descartes invent the coordinate grid system?

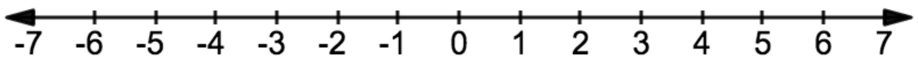
2. Name at least one way in which a coordinate system, either in two or three dimensions, is used in everyday life.

6.4.2 Exploration: Relating One-Variable and Two-Variable Inequalities

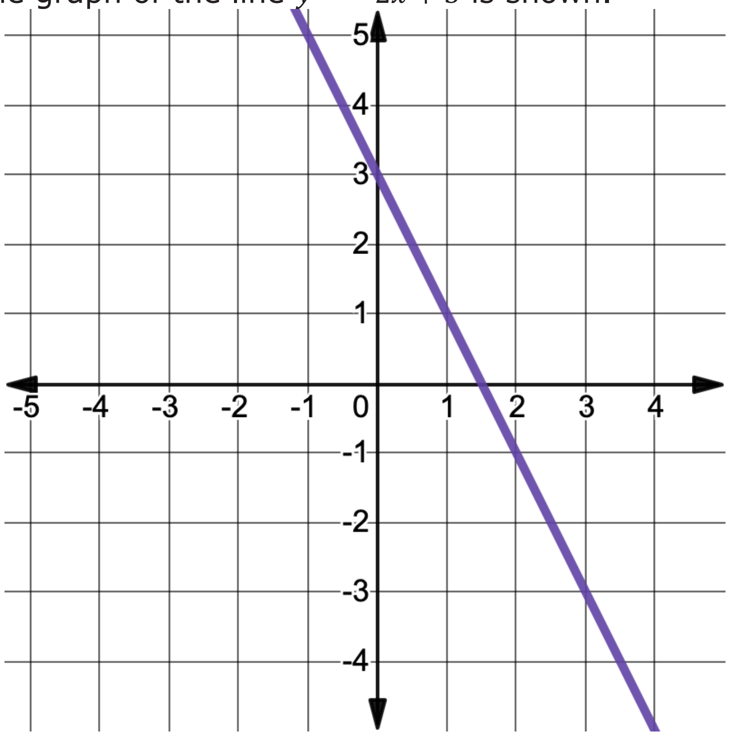
1. Use the inequality $-2x + 3 \leq 9$ to complete the following.

   a. Solve the inequality.

   b. Graph the solution set.



2. The graph of the line $y = -2x + 3$ is shown.



   a. Use either the graph or the equation to determine the y-values that complete the ordered pairs such that they satisfy the equation $y = -2x + 3$.

   $(-2, \_\_\_\_)$ $(0, \_\_\_\_)$ $(-4, \_\_\_\_)$

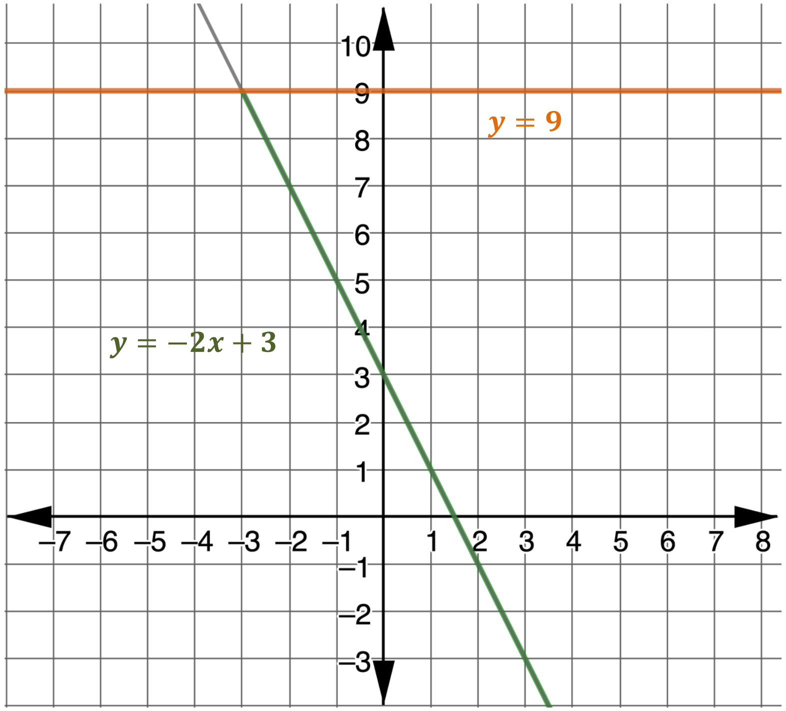
   $(-3.5, \_\_\_\_)$ $(7, \_\_\_\_)$ $(-3, \_\_\_\_)$

   b. Organize the points in the table.

Ordered Pairs With y-values Less Than or Equal to 9	Ordered Pairs With y-values Greater Than 9

   c. What do you notice about the x-values of the ordered pairs whose y-values are less than or equal to 9?

d. On the coordinate grid, the graphs of $y = -2x + 3$ and $y = 9$ are shown. The part of the graph of $y = -2x + 3$ where the y-values are less than or equal to 9 is shown in green.



For which x-values are the y-values of the equation $y = -2x + 3$ less than or equal to 9?

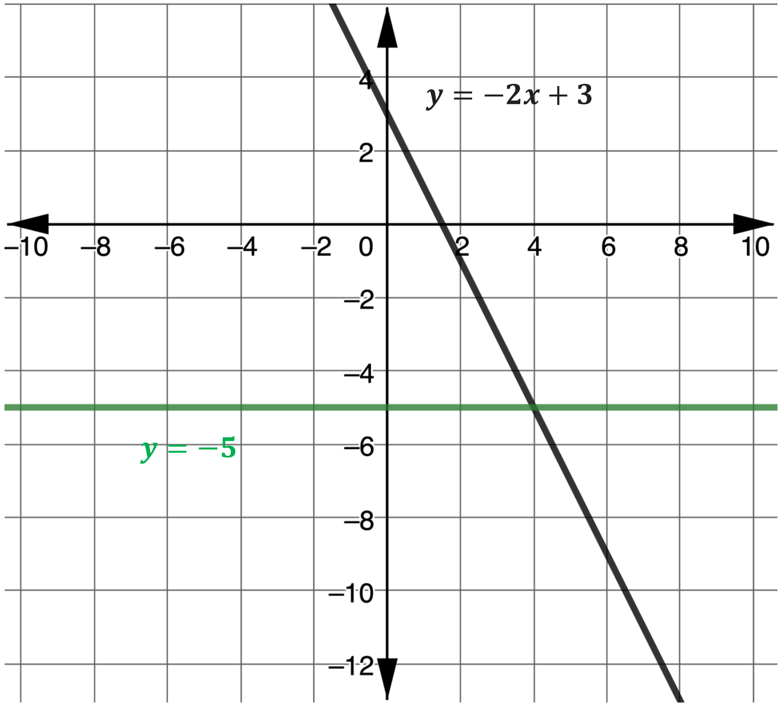
- ○ $x \geq -3$
- ○ $x \leq -3$
- ○ $x \geq 9$
- ○ $x \leq 9$

e. Explain how your answer relates to the solution set of the inequality $-2x + 3 \leq 9$.

3. Consider the inequality $-2x + 3 \leq -5$.

a. Solve the inequality.

b. The graphs of $y = -2x + 3$ and $y = -5$ are shown on the coordinate plane.



Highlight the part of the graph of $y = -2x + 3$, where the y-values are less than or equal to -5.

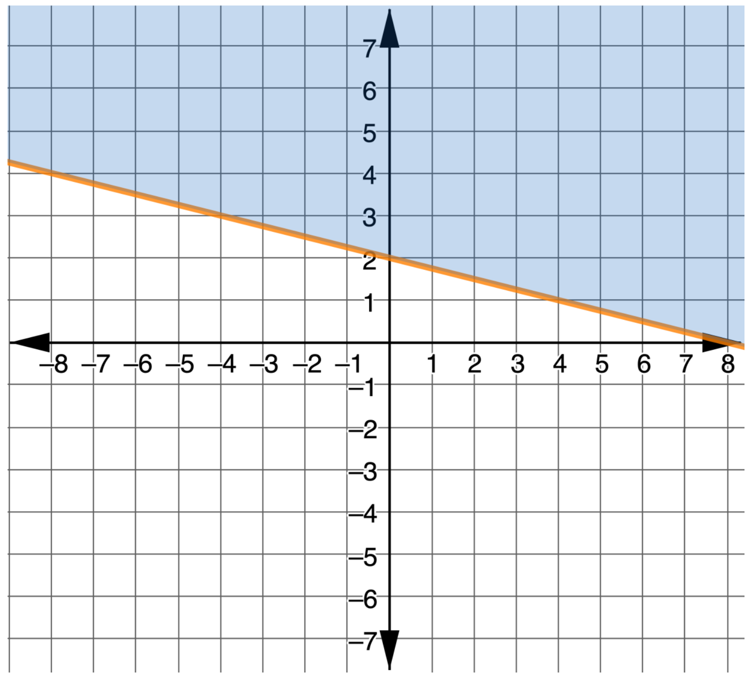
c. Describe the relationship between your solution to the inequality in Part A and the values of x for which the graph of $y = -2x + 3$ is less than or equal to -5.

4. Discuss with your partner how the graph of a line in slope-intercept form, $y = mx + b$, relates to the solution of the related one-variable inequality, $mx + b \leq k$, for some constant k. Summarize your discussion.

6.4.3 Exploration: Linear Inequalities in Two Variables

The solution sets of one-variable linear inequalities can be represented on a number line. The solution sets of two-variable linear inequalities are represented on the coordinate plane. Work with your partner to explore the graphs of two-variable linear inequalities.

1. The graph of the two-variable linear inequality $y \geq -\frac{1}{4}x + 2$ is shown.



a. The orange line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

b. How does the equation of the boundary line relate to the inequality?

c. Select one ordered pair for each of the criteria given in the table, then substitute each ordered pair into the inequality.

	An Ordered Pair in the Shaded Region	**An Ordered Pair in the Unshaded Region**	**An Ordered Pair on the Boundary Line**
Coordinates of Selected Point			
Substitute the Point into the Inequality $y \geq -\frac{1}{4}x + 2$			
Is it a solution to $y \geq -\frac{1}{4}x + 2$**?**	○ Yes ○ No	○ Yes ○ No	○ Yes ○ No

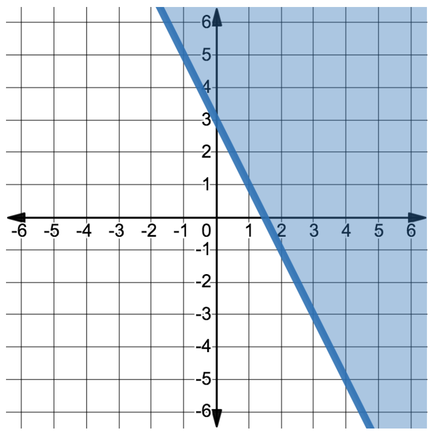
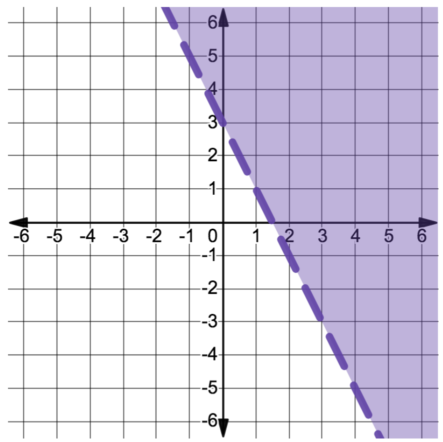
d. Describe what part(s) of the graph represents the solution set of the inequality $y \geq -\frac{1}{4}x + 2$.

e. Which part of the graph of the solution set of $y \geq -\frac{1}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	**Graphical Representation**
$y = -\frac{1}{4}x + 2$	○ boundary line ○ shaded region
$y > -\frac{1}{4}x + 2$	○ boundary line ○ shaded region

f. Discuss with your partner how you think the graph of $y \leq -\frac{1}{4}x + 2$ would look different from the graph of $y \geq -\frac{1}{4}x + 2$.

2. Two linear inequalities and their graphs are shown.

$y \geq -2x + 5$	$y > -2x + 5$
	

a. Discuss with your partner why you think the boundary line on the graph of $y \geq -2x + 5$ is solid while the boundary line on the graph of $y > -2x + 5$ is dotted. Summarize your discussion in the table.

What do you think a solid boundary line means?	What do you think a dotted boundary line means?

b. Explain how the solid and dotted lines on the graphs of two-variable inequalities relate to closed and open circles on the graphs of one-variable inequalities.

6.4.4 Bring It Together: Linear Inequalities in Two-Variables

The graph of a linear inequality in two variables includes a boundary line and shading on one side or the other side of the boundary line to represent all the ordered pairs that satisfy the inequality.

1. Complete the statements.

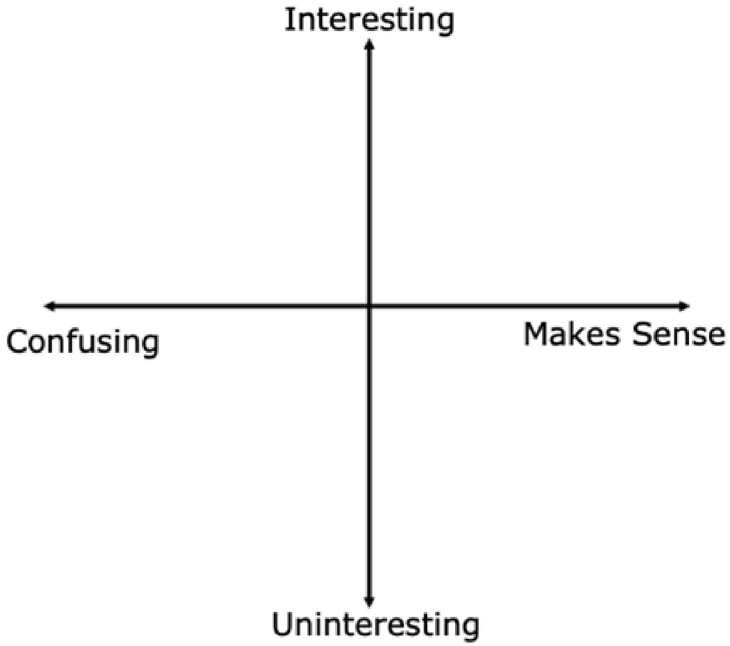
   If the points on the boundary line are included in the solution set, the boundary line is ○ dotted. ○ solid. If the points on the boundary line are not included in the solution set, the boundary line is ○ dotted. ○ solid.

On the graph of a linear inequality in two variables, the boundary line is **solid** when the expressions compared are ≥ or ≤ to each other. If they are > or <, the boundary line is **dotted**.

The other ordered pairs that are part of the solution set are represented by the shaded region. Test points on either side of the boundary line can be selected and substituted into the inequality to determine which region should be shaded to represent the solution set.

6.4.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and how much sense you made of today's learning target.



2. Explain why you chose that point and that quadrant.

Unit 6, Lesson 5: Graphing Two-Variable Linear Inequalities

Benchmark

MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

Learning Targets

- ☐ I can find the solution set of a linear inequality where one variable has a coefficient of 0.
- ☐ I can graph the solution set of a linear inequality given in standard form or point-slope form on a coordinate plane.